

# Spin–valley model uncertainty in micromagnet-assisted shuttling: a periodic-reference sensitivity atlas

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We study charge-noise dephasing and valley leakage during coherent electron spin shuttling in Si/SiGe under the synthetic spin–orbit field of a periodic micromagnet array, a controlled reference geometry motivated by proposed shuttling-bus architectures. A valley-aware co-design program would seek micromagnet geometry and shuttling trajectory that jointly suppress both loss channels, but the outcome depends on an assumed effective spin–valley coupling Hamiltonian  $H_{sv}$  that is not uniquely fixed. Rather than claiming a single robust operating basin, we report a *model-risk diagnostic*. We find that even the *existence* of a cross-ansatz robust operating point is convention-dependent: in a stray-field-only Zeeman bookkeeping the staged search finds no robust candidate surviving statistical strengthening, whereas in the physically transparent total-field convention two edge-pocket robust candidates ( $v = 10$  m/s, well closer to the magnets) survive targeted re-testing with 30 charge-noise realizations as robust in all four ansätze (4/4) at the two coupling strengths; two other apparent full-atlas candidates fail the same  $\geq 3/4$  cross-ansatz re-test. Because no generally accepted local  $\lambda_{sv}(x)$  is yet available for a shuttled Si/SiGe dot in a micromagnet field, these ansätze encode different plausible modeling assumptions rather than a derived microscopic coupling. We then quantify how strongly the coupling-ansatz choice controls the conclusions with a sensitivity atlas over 432 model-conditions. The atlas does not rank absolute shuttling performance; it ranks and classifies the *baseline-relative response* that the coupling induces in two loss channels (valley leakage and dephasing). Within this response map, the *ranking* of operating conditions is largely conserved across ansätze (mean Spearman  $\rho \approx 0.84$ ). The *thresholded classification*, which records the sign pattern of the above-threshold responses while retaining single-channel and below-threshold cases as separate categories, remains strongly model-dependent (mean exact-label agreement  $\approx 36\%$ ). Agreement is lowest for the  $B_z$ – $B_x$  and  $A$ – $B_x$  pairs, both of which set the  $\partial_x B_x$ -based ansatz against a  $\partial_x B_z$ -based one. A further test isolates the Zeeman-energy convention entering the resonance window. We recompute the atlas after replacing the stray-field-only form by the physically transparent total-field form, holding the four kernels, the condition grid, the noise model, and the seeds fixed. The mean rank correlation shifts from 0.68 to 0.84, and the run also changes which ansatz appears anomalous and whether any robust candidate exists at all. A bookkeeping-level choice therefore moves qualitative, design-level conclusions. We conclude that, for this reference geometry and ansatz family, calibration of the effective spin–valley coupling is a prerequisite before geometry-optimization claims can be trusted, and we provide the atlas as a reusable diagnostic. Absolute full-model error proxies must be reported separately and are not a target of this study.

## I. INTRODUCTION

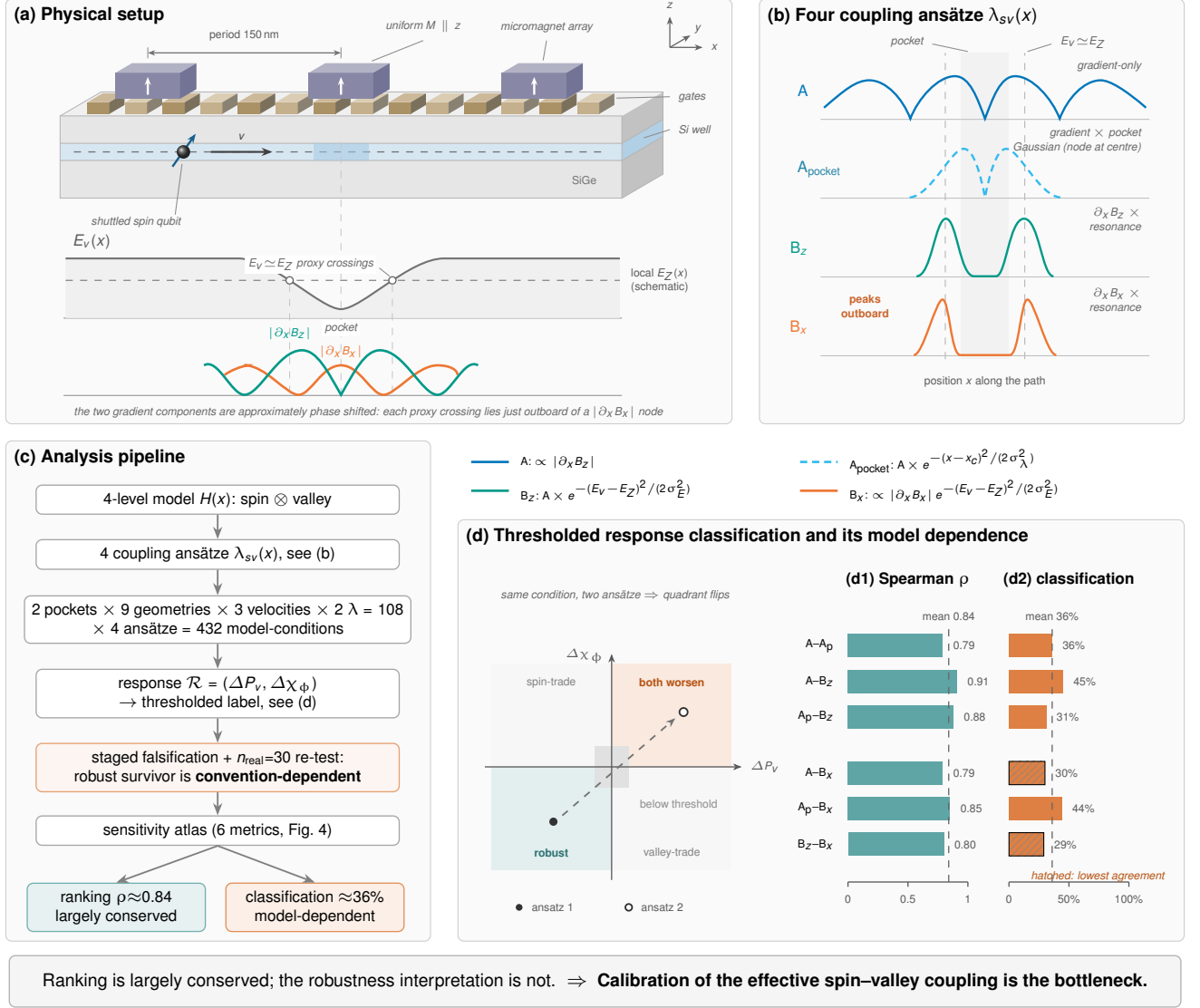
Coherent transport of electron spin qubits between distant sites (“shuttling”) is a key enabler for scalable Si/SiGe spin-qubit architectures [1–6], allowing qubit registers to be spatially separated from readout and control hardware. Silicon spin qubits have advanced rapidly, with high-fidelity single- and two-qubit gates [7–10], multi-qubit processors [11–13], quantum error correction [14], and an industrial fabrication path [15–17]; single-spin physics and readout are by now well established [18–21]. High-fidelity conveyor-mode and bucket-brigade shuttling have been demonstrated experimentally in Si/SiGe [22–24], shuttling-based two-qubit links and entanglement distribution between separated registers have been realized experimentally [10, 25, 26], and gates acting directly on moving spins have been proposed theoretically [27]. A central obstacle is the valley

degree of freedom: the near-degeneracy of the two low-lying conduction-band valleys in strained Si means that a shuttled electron can leak into the excited valley, and the spatial variation of the model valley-detuning landscape  $E_v(x)$  along the shuttling path couples motion to the valley sector; low-valley-splitting pockets arising from alloy disorder are a known threat to shuttling fidelity [28–30], and recent shuttling-based maps report physical valley splittings from 1.5 to 200  $\mu\text{eV}$  along real conveyor trajectories, dominated by alloy disorder, with trajectory selection around low-splitting regions demonstrated as a practical mitigation [31]. When combined with a micromagnet-induced synthetic spin–orbit field [32, 33], valley transitions become correlated with spin dephasing, so that geometry and trajectory choices affect spin coherence and valley leakage simultaneously.

Against this background we ask whether sensitivity-map conclusions in such architectures are stable against plausible choices of the unresolved microscopic spin–valley coupling profile. A natural program is *valley-aware co-design*: jointly optimizing micromagnet geometry and

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**FIG. 1. Overview.** (a) Device schematic: an electron spin (tilted arrow) is shuttled at constant velocity  $v$  through the Si quantum well of a Si/SiGe heterostructure by gate electrodes, beneath a periodic array of identical, uniformly  $z$ -magnetized rectangular Co prisms (period 150 nm; not to scale along  $y$ ). The periodic array provides a controlled reference geometry motivated by proposed shuttling-bus architectures. Below, aligned with the device: the diabatic valley-detuning landscape  $E_v(x)$  of Eq. (1) with a low- $E_v$  pocket, the input-detuning proxy crossings where  $E_v(x) \simeq E_Z(x) = g\mu_B[B_{\text{ext}} + B_z(x)]$  (circles), and the gradient components  $|\partial_x B_z|$  and  $|\partial_x B_x|$ , which are approximately phase shifted in space. At the array-symmetry point  $x = 0$  the former vanishes and the latter is maximal; the centred-pocket case places  $x_c$  there, so the gradient node coincides with the pocket bottom by placement rather than by necessity (Sec. III). Panel (a) draws  $E_Z$  as a horizontal level for clarity; the calculations use the full local total-field Zeeman energy of Fig. 2. (b) Four phenomenological coupling ansätze  $\lambda_{sv}(x)$  on a common position axis; the grey band marks the pocket and the dashed lines the  $E_v \simeq E_Z$  proxy crossings. For a centred pocket the resonance ansätze ( $B_z$ ,  $B_x$ ) activate symmetrically on both flanks; the  $B_x$ -gradient ansatz peaks slightly outboard of the  $\partial_x B_z$ -based ones, not on a different flank. Curves are independently height-normalized, but their horizontal peak positions are those of the released profiles. The two-line legend below the upper row gives the four kernels. (c) Analysis pipeline: the four-level spin  $\otimes$  valley model is combined with each ansatz on a common grid (432 model-conditions); a staged falsification followed by a re-test with 30 charge-noise realizations ( $n_{\text{real}} = 30$ ) shows that whether a cross-ansatz robust operating point survives is convention-dependent, and the sensitivity atlas yields two diagnostics. (d) Thresholded response classification of  $\mathcal{R} = (\Delta P_v, \Delta \chi_\phi)$  with a below-threshold zone (grey): the same operating condition can flip quadrant between ansätze. The left schematic shows only the four two-channel quadrants, whereas the 36% bar is exact agreement of the full nine-category label. (d1),(d2) Pairwise Spearman rank correlation and classification agreement for the six ansatz pairs: the magnitude ranking is largely conserved across all pairs (mean  $\rho \approx 0.84$ ) while the thresholded interpretation remains model-dependent (mean  $\approx 36\%$ , lowest for the  $B_z/B_x$  and  $A/B_x$  pairs, hatched). Together this identifies calibration of the effective spin-valley coupling as a prerequisite before geometry-optimization claims can be trusted.

shuttling trajectory to find a robust operating regime that suppresses both valley leakage and dephasing. Implicit in such optimization is a model of how the spin and valley sectors couple—an effective spin–valley Hamiltonian  $H_{sv}$ . The literature motivates several ingredients for such a coupling: gradient-induced synthetic SOC, valley-pocket localization, and spin–valley hot spots. It does not, however, provide a unique local  $\lambda_{sv}(x)$  profile for the shuttled-dot setting.

In this work we ask a prior question: *how much do co-design conclusions depend on the choice of effective coupling model?* Figure 1 summarizes the setup, the four coupling ansätze, the analysis pipeline, and the two headline diagnostics. We proceed in two steps rather than reporting a single optimized operating point. First, a staged falsification chain and a targeted higher-statistics re-test show that whether a cross-ansatz robust operating point exists *at all* depends on the modeling convention. No candidate survives in the stray-field book-keeping, whereas two grid-local edge-pocket robust candidates survive re-testing at  $n_{\text{real}} = 30$  across all four ansätze in the adopted total-field convention, where  $n_{\text{real}}$  is the number of independent charge-noise realizations averaged per condition. Second, we quantify the residual model dependence with a sensitivity atlas. Our central finding is a separation between two notions of model dependence: the *ranking* of operating conditions is largely conserved across ansätze, whereas the *physical interpretation* (which thresholded response category each condition receives) remains strongly model-dependent, with the lowest agreement in the  $B_z$ – $B_x$  and  $A$ – $B_x$  pairs. A further finding of the same kind arises one level below the ansatz choice. Correcting the Zeeman-energy convention entering the resonance window alone moves the headline statistics substantially. Modeling *conventions*, and not only ansatz selection, therefore change design-level conclusions.

## II. RELATED WORK AND POSITIONING

Prior studies of valley physics in Si/SiGe have characterized valley splitting statistics [28, 34], the role of interface disorder and valley–orbit coupling [35, 36], and shuttling-induced valley excitation [2, 29, 37, 38]. Work on micromagnet-based spin manipulation [32, 33] has established synthetic spin–orbit coupling as a control resource, and the role of spin–orbit interaction and  $g$ -tensor anisotropy in quantum dots has been studied in both electron and hole systems [39–42]. Charge-noise dephasing during shuttling and in static dots has been modeled and measured with filter-function and disorder-aware approaches [2, 19, 43–45], and hopping/baseband control with site-dependent quantization axes has emerged as an alternative to resonant driving [32, 46]. Disorder and strain effects on spin–orbit coupling have been quantified in related Ge hole-spin device models [47, 48]. Trajectory and device parameters for shuttling *have* been optimized

in specific studies, including velocity and design trade-offs in blueprint analyses and disorder-aware shuttling strategies [2, 29]. Generic optimal-control machinery is also available [49, 50].

In such studies the effective coupling model is held fixed, which is the right choice where the model is validated for the regime in question. The setting we address is the local spin–valley coupling of a dot shuttled through a micromagnet field. No generally validated local  $\lambda_{sv}(x)$  that can be adopted without additional modeling assumptions is yet available for this setting (Sec. III), so there the model choice itself becomes a source of design risk. Quantifying that risk, rather than optimizing under any one choice, is the purpose of this work. Recent conveyor-mode measurements increasingly constrain the relevant microscopic ingredients. They provide spatial maps of the valley splitting along shuttling trajectories and, together with valley-resolved  $g$ -factor maps, constrain the complex intervalley coupling [31, 51]. These measurements do not by themselves determine the effective local moving-dot spin–valley kernel parameterized here. They sharpen the experimental target that a future microscopic  $\lambda_{sv}(x)$  must fit. Valley-aware co-design in the sense used here means jointly tuning geometry and trajectory against both valley leakage and dephasing. It is accordingly a prospective program rather than an established literature.

Our contribution is orthogonal and methodological. Instead of optimizing under one model, we benchmark four spatial-profile ansätze for  $\lambda_{sv}(x)$  against each other and quantify the sensitivity of the resulting conclusions. Each ansatz is a phenomenological kernel multiplying one fixed minimal spin–valley operator. Micromagnets and electron shuttling are individually established experimentally [22, 23, 32, 33], but periodically patterned micromagnet shuttling architectures remain largely proposed or emerging designs [1, 2, 32]; we accordingly treat the periodic array as a controlled reference geometry for the diagnostic comparison. Whether a robust candidate exists is itself convention- and ansatz-dependent. That result and the diagnostic atlas together identify the next bottleneck as calibration of the effective spin–valley coupling model, not another pass of geometry optimization alone.

## III. MODEL

We consider an electron shuttled at constant velocity  $v$  along  $x$  through a periodic micromagnet array, with a position-dependent diabatic valley detuning

$$E_v(x) = E_{v,\text{base}} - (E_{v,\text{base}} - E_{v,\text{min}}) \exp\left[-\frac{(x - x_c)^2}{2\sigma_p^2}\right], \quad (1)$$

a baseline with a Gaussian pocket (minimum) near the shuttling path, using  $E_{v,\text{base}} = 100 \mu\text{eV}$ ,  $E_{v,\text{min}} = 5 \mu\text{eV}$  and  $\sigma_p = 30 \text{ nm}$ ; for  $A_{\text{pocket}}$  we set  $\sigma_\lambda = \sigma_p$ . The

TABLE I. Representative simulation parameters and their physical motivation. Values are chosen to be representative of Si/SiGe micromagnet-shuttling devices; the study varies geometry and trajectory around these baselines rather than fitting a specific device. The valley-energy scales are motivated by measured Si/SiGe valley splittings [28, 34] and by conveyor-trajectory maps [31, 37]. Literature citations identify the experimental or design scales that motivate each choice; the individual entries are not copied from the cited devices, and the reference geometry is checked independently against micromagnet field and gradient scales in the Supplementary Material. The simulator’s numerical consistency and its reproduction of the relevant literature limits are documented in the Supplementary Material.

| parameter                           | value                 | motivation / representative scale               |
|-------------------------------------|-----------------------|-------------------------------------------------|
| micromagnet period                  | 150 nm                | controlled-reference period (see caption)       |
| <code>half_x/half_z</code>          | 25/15 nm              | $50 \times 50 \times 30$ nm reference prism     |
| saturation $M_s$                    | $1.4 \times 10^6$ A/m | Co-like model value                             |
| external field $B_{\text{ext}}$     | 0.5 T                 | chosen operating-field baseline; enters Eq. (3) |
| $E_{v,\text{base}}$ far from pocket | 100 $\mu\text{eV}$    | measured splitting scales                       |
| $E_{v,\text{min}}$ at pocket centre | 5 $\mu\text{eV}$      | pocket floor, measured range                    |
| pocket width $\sigma_p$             | 30 nm                 | sets path span $8\sigma_p$                      |
| valley off-diagonal $\Delta_v$      | 0.5 $\mu\text{eV}$    | avoided-crossing scale (toy)                    |
| resonance width $\sigma_E$          | 10 $\mu\text{eV}$     | regularized hot-spot window (toy proxy)         |
| shuttling velocity $v$              | 5–20 m/s              | conveyor-mode sweep [2, 23]                     |

constant-velocity trajectory spans  $x \in [x_c - 4\sigma_p, x_c + 4\sigma_p]$ , a total path length of 240 nm. We use this detuning landscape as a phenomenological surrogate for the measured spatial variation of the physical valley splitting, which is expected from alloy-disorder and interface physics [2, 28, 34], and has recently been mapped directly along a shuttling path by conveyor-mode spin-coherent measurements [37]. Representative values for the device geometry, valley energetics, and shuttling velocity are summarized in Table I.

We model the spin and valley degrees of freedom with a minimal four-level effective Hamiltonian

$$H(x) = \frac{1}{2}g\mu_B[B_{\text{ext}} + B_z(x)]\sigma_z \otimes I_v + \frac{1}{2}g\mu_BB_x(x)\sigma_x \otimes I_v + \frac{1}{2}\varepsilon_v(x)I_s \otimes \tau_z + \Delta_v I_s \otimes \tau_x + \lambda_{sv}(x)\sigma_x \otimes \tau_x, \quad (2)$$

where  $\sigma_i$  and  $\tau_i$  are Pauli operators on the spin and valley two-level spaces,  $B_z(x)$  and  $B_x(x)$  are the longitudinal and transverse components of the *micromagnet stray field*, and  $B_{\text{ext}} = 0.5$  T is the uniform external field along  $z$ . The spin block is the Zeeman Hamiltonian in the total longitudinal field  $B_{\text{ext}} + B_z(x)$  together with the micromagnet transverse field; the transverse term  $g\mu_BB_x(x)$  is a position-dependent transverse Zeeman *energy*, not an AC EDSR drive. Because  $B_{\text{ext}}$  is spatially uniform it drops out of every spatial derivative, so the gradient-based ansätze below use the micromagnet stray-field gradients  $\partial_x B_z(x)$  and  $\partial_x B_x(x)$ . The valley block is a generic two-level avoided-crossing model with diabatic detuning  $\varepsilon_v(x)$  and off-diagonal coupling  $\Delta_v$ , so its isolated eigenlevel splitting is  $E_{\text{gap}}(x) = \sqrt{\varepsilon_v(x)^2 + 4\Delta_v^2}$ . Over the 5–100  $\mu\text{eV}$  input range used here this exceeds  $\varepsilon_v(x)$  by less than 2%, with the largest relative difference at the pocket minimum and less than 0.02% near the  $E_v \simeq E_Z$  crossings that set the proxy hot spots. The diabatic input  $\varepsilon_v(x)$  is the landscape  $E_v(x)$  of Eq. (1), whose symbol we keep to match the released code and data;

it should therefore not be read as the dressed eigenlevel gap. We treat  $\Delta_v$  as a phenomenological mixing parameter; microscopically, such intervalley coupling arises from interface and valley-orbit physics [35]. This Hamiltonian is a diagnostic effective model, not a microscopically derived Si/SiGe  $H_{sv}$ : the term  $\lambda_{sv}\sigma_x \otimes \tau_x$  is a minimal real spin-flip/valley-flip Pauli-product ansatz (one of several possible structures, e.g.  $\sigma_x\tau_y$  or  $\sigma_y\tau_x$ ), and we do not claim it to be a unique microscopic form. All four ansätze below share this single operator structure and differ only in the scalar spatial profile  $\lambda_{sv}(x)$ : we compare four phenomenological spatial kernels multiplying one fixed minimal operator, not four microscopic Hamiltonians. Microscopically, spin-valley mixing follows from matrix elements of intrinsic and synthetic spin-orbit terms between valley-dependent envelope states [36, 52].

The coupling strength is set by a profile  $\lambda_{sv}(x)$  whose form is itself uncertain. Rather than assume a single profile, we compare four *phenomenological* toy-level ansätze. These are motivated by known ingredients but are not derived from them (spatial Zeeman-energy variation from the micromagnet gradients [2, 43], transverse-gradient synthetic spin-orbit coupling [36, 53, 54], valley-pocket localization [28, 29, 37], and the physical spin-valley hot spot associated with near-degeneracy of the dressed spin and valley sectors [52, 55], which the released kernels represent by the input-detuning proxy  $E_v(x) \simeq E_Z(x)$ ). We use the physically transparent total-field convention throughout, in which the Zeeman energy is evaluated with the full local field,

$$E_Z(x) = g\mu_B [B_{\text{ext}} + B_z(x)], \quad (3)$$



with  $B_{\text{ext}} = 0.5 \text{ T}$  (Table I). The four ansätze are then

$$\text{A: } \lambda_{sv}(x) = \lambda_0 \frac{|\partial_x B_z(x)|}{\max_x |\partial_x B_z|}, \quad (4)$$

$$\text{A}_{\text{pocket}}: \lambda_{sv}(x) = \lambda_{sv}^{\text{A}}(x) \exp\left[-\frac{(x - x_c)^2}{2\sigma_\lambda^2}\right], \quad (5)$$

$$\text{B}_z: \lambda_{sv}(x) = \lambda_{sv}^{\text{A}}(x) \exp\left[-\frac{(E_v(x) - E_Z(x))^2}{2\sigma_E^2}\right], \quad (6)$$

$$\text{B}_x: \lambda_{sv}(x) = \lambda_0 \frac{|\partial_x B_x(x)|}{\max_x |\partial_x B_x|} \exp\left[-\frac{(E_v(x) - E_Z(x))^2}{2\sigma_E^2}\right], \quad (7)$$

where  $E_Z(x)$  is the local Zeeman energy of Eq. (3), so that the diagnostic condition  $E_v(x) \simeq E_Z(x)$  produces two flank proxy crossings around the pocket. (Under an alternative bookkeeping convention, the window is evaluated with a stray-field-only Zeeman energy, which shifts the resonance windows toward the pocket bottom and, as we show in Sec. V, moves the atlas statistics substantially. This convention sensitivity is itself part of the model risk we report.)

A is a longitudinal-gradient diagnostic baseline, motivated by the spatial Zeeman-energy variation the micromagnet imprints rather than by a microscopic spin-flip coupling.  $\text{A}_{\text{pocket}}$  is a spatial-localization ansatz that weights the gradient coupling by a Gaussian envelope centred on the low- $E_v$  pocket.  $\text{B}_z$  is a longitudinal-gradient resonance stress test rather than a literature-derived spin-valley matrix element.  $\text{B}_z$  and  $\text{B}_x$  are resonance-window ansätze motivated by the spin-valley hot spot [52, 55]. Their Gaussian windows are centred on the input-detuning condition  $E_v(x) \simeq E_Z(x)$  and are diagnostic proxies for the dressed crossing rather than microscopic resonance line shapes. They use two alternative choices of magnetic-gradient component: the longitudinal gradient  $\partial_x B_z$  sets the local Zeeman-splitting variation (the dephasing gradient [2, 43]), whereas the transverse gradient  $\partial_x B_x$  is the component that drives spin flips in slanting-field EDSR [54, 56, 57]. Neither choice is prescribed for a shuttled dot by the literature, which is precisely the model uncertainty probed here.

None of the four profiles should be interpreted as a literature-standard or microscopically derived spin-valley coupling; the Gaussian resonance window  $e^{-(E_v - E_Z)^2 / (2\sigma_E^2)}$  in particular is a regularized diagnostic proxy for the hot-spot enhancement, not a standard expression. They are diagnostic kernels used to quantify how strongly the conclusions depend on plausible, non-unique modeling choices.

One might ask whether the gradient-only kernels A and  $\text{A}_{\text{pocket}}$  are physically admissible, given the known spin-valley enhancement near  $E_v \approx E_Z$ . In the diabatic basis of the Hamiltonian above, the operator  $\lambda_{sv}(x) \sigma_x \otimes \tau_x$  couples a pair of states whose uncoupled energy difference is  $E_v(x) - E_Z(x)$ . A nonzero off-diagonal coupling at a flank crossing therefore produces an avoided crossing and resonantly enhanced spin-valley mixing through the four-

level dynamics, even when  $\lambda_{sv}(x)$  contains no explicit resonance window [52, 55].

For the profiles considered here, A and  $\text{A}_{\text{pocket}}$  remain nonzero at the relevant flank crossings. The window in  $\text{B}_z/\text{B}_x$  tests the distinct, stronger phenomenological assumption that the effective off-diagonal coupling amplitude is itself localized near resonance, thereby adding profile localization to the dynamical level-crossing enhancement. Microscopic spin-valley matrix elements depend on spin-orbit and field-gradient matrix elements between valley-dependent envelope states and need not take the particular Gaussian form in  $E_v - E_Z$  used here [36, 52]. The A-type kernels should therefore be read as diagnostic detuning-independent profiles, not as uniquely derived microscopic matrix elements. Neither family is a null straw man; together they bracket complementary assumptions about the spatial structure of the same off-diagonal coupling.

Figure 2 shows the four profiles for a representative pocket: the gradient-only model A is delocalized over the whole period,  $\text{A}_{\text{pocket}}$  concentrates the gradient coupling near the pocket, and the resonance models  $\text{B}_z/\text{B}_x$  are sharply peaked where  $E_v(x)$  crosses  $E_Z(x)$ . An important subtlety arises because all three of  $\text{A}_{\text{pocket}}$ ,  $\text{B}_z$ , and  $\text{B}_x$  *multiply the field gradient*  $|\partial_x B_z|$  (or  $|\partial_x B_x|$ ). The two gradient components are set by the array alone, not by the pocket. By symmetry  $\text{B}_z$  is extremal and  $\text{B}_x$  vanishes at the array-symmetry point  $x = 0$ , so  $|\partial_x B_z|$  has a node there; for the fitted baseline field  $|\partial_x B_x|$  also attains its maximum at that point. The centred-pocket configuration places  $x_c = 0$  there as well, so the node-pocket alignment is a placement choice rather than a generic property of a centred valley pocket. Consequently  $\text{A}_{\text{pocket}} = |\partial_x B_z| \times \text{Gaussian}$  is *zero at the pocket centre* and forms two symmetric peaks on the flanks rather than a single peak at the centre.  $\text{A}_{\text{pocket}}$  is thus “pocket-localized” in its Gaussian envelope but inherits the gradient’s flank structure; only when the pocket sits off-centre (where the gradient no longer vanishes at the well bottom) does its weight become strongly asymmetric and single-flank dominated. The resonance windows ( $\text{B}_z$ ,  $\text{B}_x$ ) activate symmetrically on both flanks for the centred pocket as well; the profiles therefore differ not in which side of the pocket they occupy but in their *shape, width, phase relative to the driving gradient, and amplitude*. In particular, because  $\text{B}_x$  is built from  $\partial_x B_x$  rather than  $\partial_x B_z$ , and the two gradient components are approximately phase shifted, its weight peaks slightly outside the crossing where the  $\partial_x B_z$ -based kernels peak slightly inside it. This offset is consistent with the distinct thresholded-classification behavior that  $\text{B}_x$  shows in the atlas. (A left/right asymmetry appears only for an off-centred pocket, where the two flank crossings are inequivalent; the centred-pocket profiles used here are two-sided and symmetric, and any single-sided “opposite side” reading of them is an argmax tie-break artifact of exactly degenerate twin peaks.)

We emphasize that  $\Delta_v$  and  $\sigma_E$  are toy parameters of

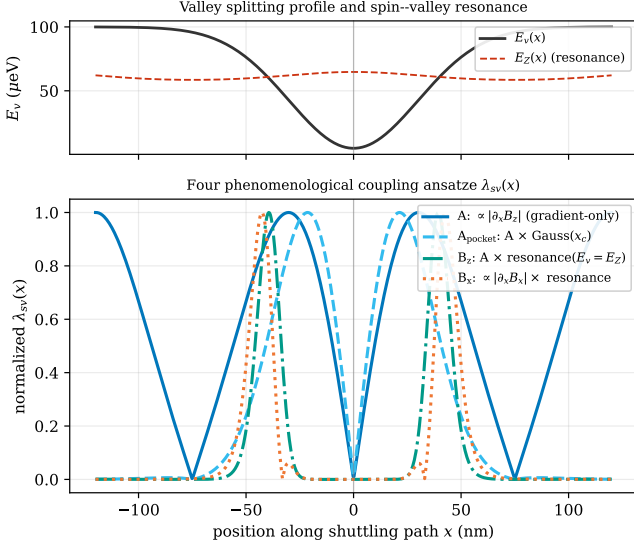


FIG. 2. Diabatic valley-detuning landscape  $E_v(x)$  with a pocket (top) and the four phenomenological coupling ansätze  $\lambda_{sv}(x)$  (bottom), normalized to unit peak. All three of  $A_{\text{pocket}}$ ,  $B_z$ , and  $B_x$  form symmetric twin peaks on the pocket flanks, but for two different reasons.  $A_{\text{pocket}}$  and  $B_z$  multiply the longitudinal gradient  $|\partial_x B_z|$ , whose node at the array-symmetry point coincides with the centred pocket bottom by placement, so  $A_{\text{pocket}}$  is zero at the pocket centre despite its “pocket-localized” Gaussian envelope.  $B_x$  multiplies  $|\partial_x B_x|$ , which is instead maximal at the pocket centre; its twin-peak topology therefore arises from the resonance window rather than from a central gradient node, since  $B_z$  and  $B_x$  activate only where  $E_v(x) \approx E_Z(x)$  in the total-field convention. The two gradient components are approximately phase shifted, so at the crossing  $|\partial_x B_z|$  is already past its maximum while  $|\partial_x B_x|$  is rising out of its node;  $B_z$  therefore peaks slightly inside the crossing and  $B_x$  slightly outside it. That offset, rather than a difference of flank, is how the choice of gradient component enters. The archived top-panel title uses the earlier phrase “valley splitting profile”; the plotted  $E_v(x)$  is the diabatic input detuning of Eq. (1), and “resonance” in the internal legend denotes the input-detuning window  $E_v \simeq E_Z$  rather than the dressed eigenlevel gap. For shape comparison only, each displayed curve is independently normalized to unit peak; the adopted prefactor convention retains the unequal final amplitudes in the simulations.

the diagnostic model rather than measured constants; the geometric and velocity baselines, by contrast, are representative of reported Si/SiGe shuttling devices.

The observables target the two loss channels that the literature identifies as central to Si/SiGe shuttling: valley excitation (leakage) and valley-induced spin dephasing [22, 29, 37, 38]. We quantify them as the change in valley-leakage probability  $\Delta P_v$  and the change in dephasing phase variance  $\Delta \chi_\phi$  induced by the spin-valley coupling. We use three nested models: a spin-only reference with charge noise but no valley sector (M1), a valley-aware reference without charge noise (M1V), and the full model with both charge noise and valley cou-

pling (M2). The two responses are isolated against *different* references by design, each separating one cross-effect:  $\Delta P_v = P_v^{\text{M2}} - P_v^{\text{M1V}}$  (the effect of charge noise on valley leakage) and  $\Delta \chi_\phi = \chi^{\text{M2}} - \chi^{\text{M1}}$  (the effect of the valley sector on charge-noise dephasing); the dephasing is computed within a filter-function description of  $1/f$  charge noise [19, 43, 58]. Thus  $\mathcal{R} = (\Delta P_v, \Delta \chi_\phi)$  is not a decomposition of a single infidelity but a pair of cross-effect measures.  $\Delta P_v$  and  $\Delta \chi_\phi$  are *baseline-relative response diagnostics*, not absolute error rates. Absolute performance would be read from the full-model values  $P_v^{\text{M2}}$  and  $\chi_\phi^{\text{M2}}$  (or a derived phase-error proxy  $p_\phi = (1 - e^{-\chi_\phi/2})/2$ ), which we do not optimize here.

Under the standard assumption that each component is monotonic in the infidelity of its channel (larger leakage and larger dephasing both reduce fidelity), we classify each condition by the sign of each *above-threshold* response. A condition is *robust* when both responses exceed their channel floors and improve ( $\Delta P_v < 0$  and  $\Delta \chi_\phi < 0$ ), *valley-trade* or *spin-trade* when both exceed their floors with opposite signs, and *both-worsen* when both exceed their floors and worsen, using effect-size floors  $|\Delta P_v| \geq 10^{-4}$ ,  $|\Delta \chi_\phi| \geq 10^{-3}$ . The realization count  $n_{\text{real}}$  introduced above counts Monte Carlo repetitions of the simulated noise, not experimental repetitions, and is the column name used in the released data tables and scripts. These floors are not taken from an external fidelity standard, and they are not statistical-significance cutoffs or Monte Carlo confidence bounds. They are operational criteria that define a dead zone on channel-appropriate response scales; the decade offset between  $10^{-4}$  and  $10^{-3}$  reflects that  $|\Delta \chi_\phi|$  is typically about an order of magnitude larger than  $|\Delta P_v|$  in our data. Two separate checks probe different sensitivities. A threshold sweep in the Supplementary Material tests dependence on the floor choice. Candidate stability is assessed separately by re-running the four screened  $\geq 3/4$  candidates at  $n_{\text{real}} = 30$  (Sec. V), which is where individual low-statistics classifications are tested. When only one channel is above threshold, the condition is placed in a separate single-channel category rather than forced into a two-channel quadrant; this avoids contaminating the *both-worsen* class with one-channel cases. The classification uses only the signs of the above-threshold responses and is therefore insensitive to their relative scale; the rank correlation, by contrast, uses the response magnitude  $|\mathcal{R}| = \sqrt{\Delta P_v^2 + \Delta \chi_\phi^2}$ , whose ordering is preserved under uniform rescaling but does depend on the relative weighting of the two channels (we report this dependence as a robustness check).

A co-design decision reads off the direction of the channel responses (whether each channel improves or worsens), not only their magnitude. We therefore treat the four two-channel quadrants as the decision-facing sign diagnostic and the ranking as a complementary magnitude-ordering diagnostic. The ranking remains informative, but the two diagnostics answer different questions and exhibit different degrees of model dependence.

## IV. METHODS

### A. A staged falsification chain

A central methodological point is that we do not assert a robust basin from small-sample evidence. We first ran a six-stage verification chain, in the original stray-field convention, in which each stage either strengthens statistics or adds a coupling model, and we tracked whether cross-validated robust candidates survive (Table II). In that convention no candidate survived; as we show below, however, the outcome of this same procedure *changes* under the adopted total-field convention, which is itself one of our results.

The stage-5 re-test raises the statistics to  $n_{\text{real}} = 30$ . Every candidate identified in the low-statistics stages ( $n_{\text{real}} = 2\text{--}5$ ) then fails the criterion *in the stray-field convention*: none of the earlier detections survives the strengthened statistics. Extending to the four-ansätze family (stage 6; A,  $A_{\text{pocket}}$ ,  $B_z$ ,  $B_x$ ) evaluates 72 model rows; among the 12 positive- $\lambda$  physical settings, none is robust in  $\geq 3/4$  ansätze in that convention. This staged search motivated reframing the study from basin discovery to a coupling-ansatz sensitivity atlas. As we show in Sec. V, repeating the  $n_{\text{real}} = 30$  re-test in the adopted total-field convention reverses the stage-5/6 conclusion for two grid-local candidates. We therefore present the atlas, together with the convention- and ansatz-dependence of the robust/fragile verdict, rather than a single basin claim.

### B. Sensitivity atlas

We use two pocket placements:  $x_c = 0$  for the centred case, aligned with the array-symmetry point, and  $x_c = 25$  nm for the off-centred edge case. On a grid of geometries (nine unique points: the baseline plus one additive + and − step in each of four geometry parameters, **period** ( $\pm 30$  nm), **depth** ( $\pm 10$  nm), **half\_x** ( $\pm 5$  nm), and **half\_z** ( $\pm 3$  nm), i.e.  $1 + 4 \times 2 = 9$ ), velocities, and coupling strengths  $\lambda$ , we compute six metrics. Here **depth** is the signed vertical coordinate of the quantum well relative to the magnet plane,  $-50$  nm at baseline; a + step moves the well toward the magnet array, reducing the magnet–well separation and strengthening the stray field. The metrics are (i) per-model response amplitude; (ii) raw geometry sensitivity  $S_\theta = |\mathcal{R}(\theta + \Delta\theta) - \mathcal{R}(\theta - \Delta\theta)|/2\Delta\theta$ ; (iii) model distance  $D_{m_1 m_2}$  (raw and channel-balanced); (iv) pairwise Spearman rank correlation of  $|\mathcal{R}|$ ; (v) classification agreement, the fraction of conditions assigned the same thresholded response label, drawn from nine categories: the four two-channel quadrants (*robust*, *valley-trade*, *spin-trade*, *both-worsen*), four single-channel categories, and *below-threshold*. We reserve “quadrant” for the four two-channel labels; the released pipeline and the Fig. 4 panel titles retain the shorthand “quadrant agreement” for this metric; (vi) relative geometry sensitivity

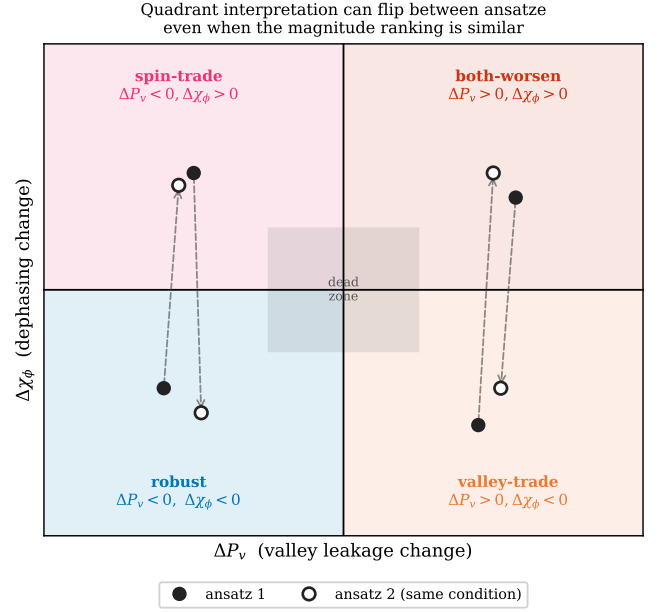


FIG. 3. Quadrant classification of the response  $\mathcal{R} = (\Delta P_v, \Delta \chi_\phi)$  into robust, valley-trade, spin-trade, and both-worsen, with an effect-size dead zone. The panel is a schematic and carries no data axes: the filled and open markers illustrate one operating condition evaluated under two coupling ansätze. The two responses can have similar magnitudes, so their ranking is preserved. Their assigned quadrants can nevertheless differ, which is why classification agreement is the more model-dependent quantity. Computed examples of such quadrant flips for the  $A \rightarrow A_{\text{pocket}}$  pair on the full validation grid are shown in Fig. 4(5).

(normalized by  $\theta_0$ ). The  $\lambda = 0$  slice is retained only as an invariance check (separable limit,  $\Delta \mathcal{R} \approx 0$ ). Realizations share common random numbers across models to reduce comparison variance. The reported rank and classification diagnostics are empirical summaries over the fixed 432-condition grid.

The distinction between metrics (iv) and (v) is central to our conclusions and is illustrated in Fig. 3: two models may rank a set of conditions by response *magnitude* almost identically (high rank correlation) while still assigning many conditions to *different* categories (low classification agreement). Ranking captures “how large is the response,” whereas the thresholded label records which response signs are resolved above the stated floors. Within that label set, the four two-channel quadrants capture the sign pattern of the baseline-relative responses, which is one ingredient of a co-design decision; a final performance decision would also require the absolute full-model leakage and dephasing proxies, which we do not optimize here.

TABLE II. Staged falsification chain *in the stray-field Zeeman convention*. Each stage either strengthens statistics (more charge-noise realizations  $n_{\text{real}}$ ) or broadens the model space, and we track whether cross-validated robust candidates survive. A single condition is operationally classified as “robust” if both responses exceed their stated effect-size floors and improve,  $\Delta P_v = P_v^{\text{M2}} - P_v^{\text{M1V}} < 0$  and  $\Delta \chi_\phi = \chi^{\text{M2}} - \chi^{\text{M1}} < 0$ , after the stated effect-size thresholds (note the two responses use different references); it is “cross-robust” if it is robust in both coupling ansätze tested at that stage (A and  $A_{\text{pocket}}$ ), and “strong” if it is robust in  $\geq 3/4$  ansätze. In this convention the single candidate seen at low statistics disappears once  $n_{\text{real}}$  is increased and the model family is broadened. The “conditions” column counts model $\times$ geometry combinations in stages 1 and 6 (two and four ansätze respectively) and geometry conditions in stages 3–5 (with the two ansätze entering the cross-robust test); each entry is tracked individually. Under the adopted total-field convention this procedure yields a different outcome (Sec. V): two grid-local edge-pocket robust candidates survive.

| stage | description                        | $n_{\text{real}}$ | conditions | cross-robust | outcome                                        |
|-------|------------------------------------|-------------------|------------|--------------|------------------------------------------------|
| 1     | initial coarse scan (2 ansätze)    | 5                 | 48         | 1            | one candidate ( $A \wedge A_{\text{pocket}}$ ) |
| 2     | objective-weight robustness sweep  | –                 | 5 sets     | –            | ranking stable to weighting                    |
| 3     | local refinement around candidate  | 2                 | 9          | 1            | one neighboring condition                      |
| 4     | full local grid                    | 2                 | 45         | 2            | two low-statistics candidates                  |
| 5     | targeted higher-statistics re-test | 30                | 6          | 0            | no candidate survives                          |
| 6     | four-ansätze family re-test        | 5                 | 72         | 0            | no $\geq 3/4$ -robust condition                |

## V. RESULTS

*a. Ranking vs. interpretation.* Over the 432 model-conditions of the full validation grid (centred and edge pockets,  $v \in \{5, 10, 20\}$  m/s,  $n_{\text{real}} = 5$ ; executed as a case split and merged), the mean rank correlation is  $\rho = 0.84$  [Fig. 4(1)], with all six pairwise values in a narrow high band (0.79–0.91). Condition ranking is therefore *largely conserved* across the ansätze. In contrast, the mean classification agreement is only 36% [Fig. 4(2)], with pairwise values spanning 29%–45%: the physical interpretation (which category a condition falls into) is substantially more model-dependent than the ranking. The separation between a well-conserved magnitude ordering and a poorly-conserved thresholded classification is the central quantitative result, and it is present even though the ranking itself is high. Two checks address the two most obvious alternative explanations: skewed category marginals and the inclusion of non-quadrant labels. Chance-corrected Cohen’s  $\kappa$  stays modest for all six pairs (0.135–0.334), so the low raw agreement is not simply a consequence of skewed category marginals, of which *below-threshold* is the largest. And restricting the comparison to the conditions that both ansätze place in a two-channel quadrant — removing the below-threshold and single-channel categories entirely — leaves agreement at 44.8% (47 of 105 such comparisons). That conditional subset is small and selected, but it shows that low agreement is not confined to the excluded categories.

*b. Where the dependence lives.* The residual model dependence is not uniform across pairs, and it now lives almost entirely in the thresholded classification rather than in the ranking. The two lowest-agreement pairs both set the  $\partial_x B_x$ -based model against a  $\partial_x B_z$ -based model:  $B_z$  vs.  $B_x$  at 29% and A vs.  $B_x$  at 30%. The two  $\partial_x B_z$ -based models A and  $B_z$  agree best (45% classification agreement,  $\rho = 0.91$ ). Even for the pair with the smallest overall model distance, A vs.  $A_{\text{pocket}}$ , individual conditions still change quadrant [Fig. 4(5)].  $B_x$  is not,

however, a uniform outlier: the third  $B_x$  pair,  $A_{\text{pocket}}$  vs.  $B_x$ , is comparatively high (44%); this is consistent with the pocket-localized  $A_{\text{pocket}}$  sharing  $B_x$ ’s concentration near the pocket flanks. The decisive modeling axis is therefore the *gradient component* entering the synthetic coupling ( $\partial_x B_z$  vs.  $\partial_x B_x$ ), modulated by spatial localization, rather than the presence of a resonance window per se. Under the total-field convention the rank correlations of the  $B_x$  pairs (0.79–0.85) are comparable to the others: the legacy-convention appearance of a *ranking divergence* for  $B_x$  was itself an artifact of the stray-field-only Zeeman bookkeeping, and only the classification-level discrepancy survives the correction. This is direct evidence that the interpretive conclusion is controlled by modeling choices, here both the ansatz and the Zeeman convention, rather than by the data.

*c. Convention sensitivity as primary evidence.* The Zeeman-energy convention entering the resonance window is not a detail. Recomputing the entire atlas after replacing the stray-field-only Zeeman energy with the total-field form moves the mean rank correlation from 0.68 to 0.84 and the mean classification agreement from 29% to 36%, and it dissolves the apparent  $B_x$  ranking divergence noted above. Different equations trivially produce different numbers. What matters here is what the comparison holds fixed and what it flips. The four kernels, the 108-condition physical grid, the noise model, and every random seed are identical between the two runs. Only the bookkeeping of how the external field enters the hot-spot window changes. Yet the outputs that flip are categorical rather than marginal: which ansatz appears to be the outlier, and (below) whether any cross-ansatz robust point exists. Neither flip is predictable a priori from “the equations differ.” This one-variable comparison captures the paper’s thesis: design-level conclusions about valley-aware shuttling are sensitive to modeling assumptions that current literature does not fix. Physically, only the total-field convention corresponds to the actual experimental situation ( $B_{\text{ext}}$  is present); we therefore adopt it



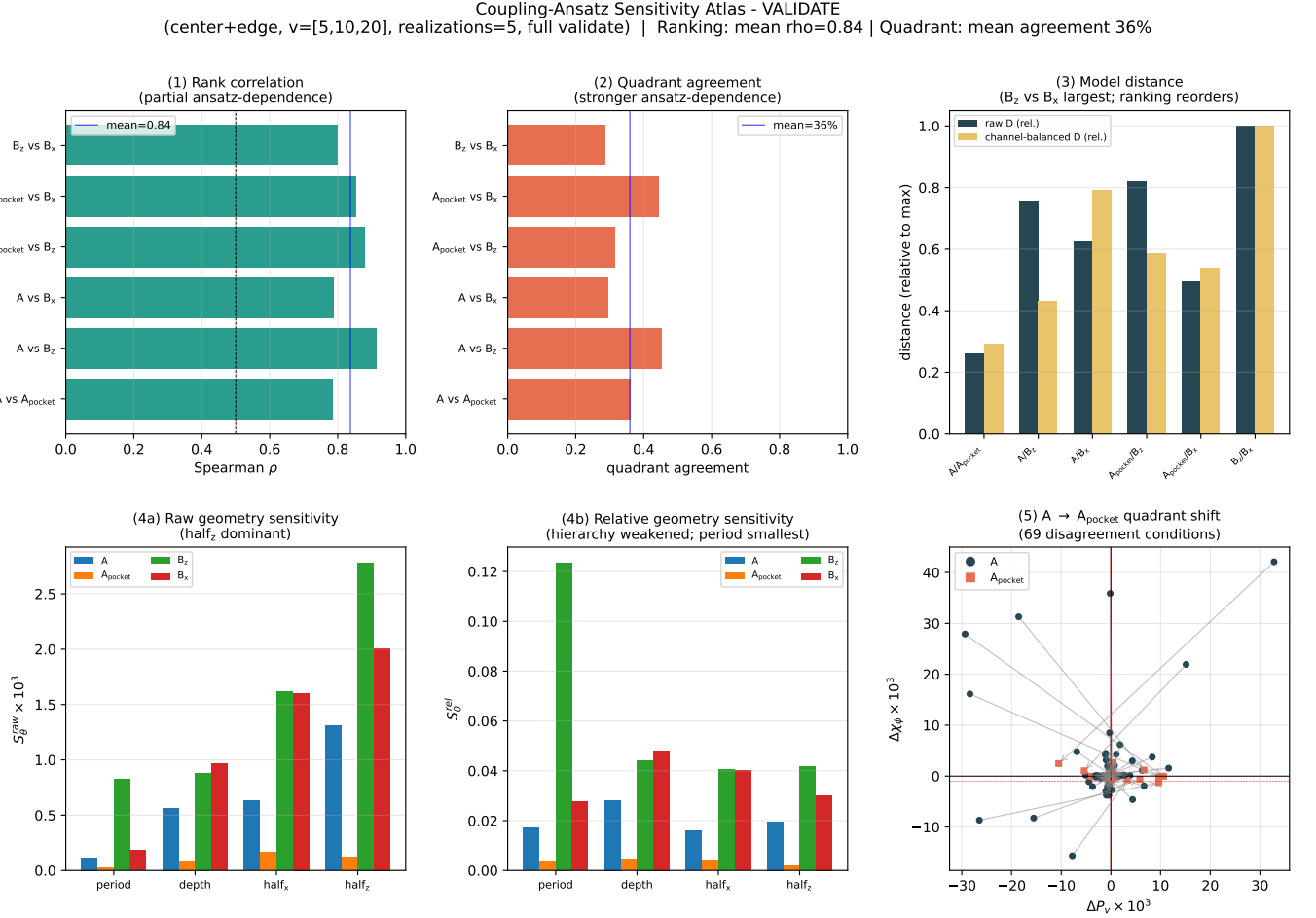


FIG. 4. Coupling-ansatz sensitivity atlas over the 432 model-conditions of the full validation grid (centred and edge pockets,  $v \in \{5, 10, 20\}$  m/s, four ansätze, two coupling strengths), each averaged over  $n_{\text{real}} = 5$  charge-noise realizations; total-field convention, Eq. (3). The panel titles retain the pipeline label “VALIDATE”, denoting this full 432-condition validation grid. (1) Pairwise Spearman rank correlation of condition magnitude (mean  $\rho = 0.84$ ). (2) Pairwise classification agreement (mean 36%), showing that thresholded labels are only modestly conserved despite the high rank-order agreement of panel (1). Panel (2), titled “quadrant agreement” by the archived generator, reports exact agreement of the full thresholded label, including the below-threshold and single-channel categories. (3) Raw and channel-balanced model distance  $D$ , largest for  $B_z$  vs.  $B_x$  (smallest for  $A$  vs.  $A_{\text{pocket}}$ ); the channel-balanced distance measures each response in units of its effect-size threshold ( $10^{-4}$  for  $\Delta P_v$ ,  $10^{-3}$  for  $\Delta\chi_\phi$ ). Because the two thresholds differ by a decade, this is a reweighting of the two channels and not a change of units: the closest pair ( $A$  vs.  $A_{\text{pocket}}$ ) and the most distant pair ( $B_z$  vs.  $B_x$ ) are unchanged, while the four intermediate pairs reorder. Each series is shown relative to its own maximum. (4a,4b) Raw and relative geometry sensitivity. Per nanometer (4a), all four models share the ordering  $\text{half}_z > \text{half}_x > \text{depth} > \text{period}$  ( $A_{\text{pocket}}$  swaps the top two). Per fractional change of each knob (4b, raw sensitivity scaled by the knob’s baseline value), no shared hierarchy survives: the most and least sensitive knobs differ between ansätze, and for  $B_z$  the *period* becomes the single most sensitive knob. (5) Individual conditions whose quadrant assignment moves between  $A$  and  $A_{\text{pocket}}$ , the pair with the smallest overall model distance  $D$  [panel (3)]. Across all pairs [panel (2)], agreement is lowest for  $B_z/B_x$  and  $A/B_x$ , each of which contrasts  $B_x$  with a  $\partial_x B_z$ -based model.

for all headline numbers and retain the stray-field run as an archived case study of how a plausible bookkeeping slip propagates to design conclusions.

*d. Convention-dependent existence of robust candidates.* The convention change also flips a stronger, discrete conclusion: whether any cross-ansatz robust operating point exists at all. In the stray-field convention the staged chain (Table II) found none surviving the strengthened statistics. In the adopted total-field atlas,

re-classifying the full  $n_{\text{real}} = 5$  grid yields four physical conditions robust in  $\geq 3/4$  ansätze (one in 4/4). We subjected exactly those four to the same targeted re-test used in stage 5, now at  $n_{\text{real}} = 30$  and in the total-field convention (Table III). Two no longer meet the  $\geq 3/4$  cross-ansatz criterion: the centred-pocket point at  $v = 5$  m/s remains robust in 1/4 ansätze and the edge point at  $v = 20$  m/s in 2/4, so their low-statistics cross-ansatz classifications do not survive. *Two survive as robust in*

TABLE III. Targeted  $n_{\text{real}} = 30$  re-test, in the adopted total-field convention, of the four conditions that are robust in  $\geq 3/4$  ansätze at  $n_{\text{real}} = 5$ . Two survive as robust in all four ansätze; the other two no longer meet the  $\geq 3/4$  cross-ansatz criterion. “robust ansätze” counts models with  $\Delta P_v < 0$  and  $\Delta\chi_\phi < 0$  above threshold.

| case   | geometry | $v$ (m/s) | $\lambda$ ( $\mu\text{eV}$ ) | robust ansätze |
|--------|----------|-----------|------------------------------|----------------|
| center | period−  | 5         | 0.5                          | 1/4            |
| edge   | depth+   | 10        | 0.5                          | 4/4            |
| edge   | depth+   | 10        | 1.0                          | 4/4            |
| edge   | depth+   | 20        | 1.0                          | 2/4            |

*all four ansätze*: the edge-pocket condition at  $v = 10$  m/s with the reduced magnet–well separation, at both coupling strengths  $\lambda \in \{0.5, 1.0\} \mu\text{eV}$ . A small grid-local cross-ansatz robust candidate set therefore does exist in the physically preferred convention. Its identification is nonetheless model-contingent: it is invisible in the stray-field bookkeeping, and even within the total-field convention two of the four screening candidates fail the higher-statistics cross-ansatz criterion. The surviving pair is itself grid-local and floor-based rather than confidence-certified: its narrowest channel clears the  $|\Delta\chi_\phi|$  floor by only about 5%. The atlas’s role is to flag which candidates warrant this higher-statistics validation. It also exposes that the robust/fragile verdict, and even the existence of a robust point, depends on the coupling convention and ansatz family, not on the geometry search alone.

*e. Geometry sensitivity.* Per nanometer of knob variation, all four models share the raw ordering **half\_z** > **half\_x** > **depth** > **period** (with  $A_{\text{pocket}}$  swapping the top two): the magnet thickness dominates and the array period is the weakest per-nm knob. This shared picture does not survive rescaling to *fractional* knob changes [Fig. 4(4b)]: the per-fraction hierarchy differs between ansätze. Depth leads for  $A$ ,  $A_{\text{pocket}}$ , and  $B_x$ , whereas for  $B_z$  the *period* becomes the single most sensitive knob; a fractional period change rescales the stray-field amplitude at the quantum-well depth roughly as  $e^{-2\pi z/a}$  and thereby shifts the resonance crossings to which  $B_z$  responds. The least-sensitive knob likewise varies. The geometry-sensitivity hierarchy is therefore itself ansatz-dependent once knob scales are accounted for, and we caution against any absolute knob-importance claim; this extends the model-risk message from the operating-condition verdicts to the geometry story itself.

## VI. DISCUSSION

The results indicate that conclusions about valley-aware geometry/trajectory co-design are sensitive to the assumed effective  $H_{sv}$ , particularly to the choice of magnetic-field gradient component and to the Zeeman-energy convention. The four two-channel quadrants form

the decision-facing sign diagnostic: they distinguish robust, trade-off, and both-worsen responses. This is also the diagnostic we find to be most model-dependent. We do not conclude that robust operating points cannot exist: in the physically preferred total-field convention two grid-local edge-pocket candidates survive  $n_{\text{real}} = 30$  re-testing across all four ansätze. The *identification* of such a point is instead model-contingent: the same staged procedure returns “no robust candidate” in one convention and “a surviving robust candidate” in another. Robust-region claims built on a single toy coupling and a single convention are therefore not trustworthy without cross-checks.

The thresholded classification verdict is strongly coupling-dependent (mean raw cross-ansatz classification agreement  $\approx 36\%$ ; pairwise Cohen’s  $\kappa = 0.135\text{--}0.334$  on the same label set), and this dependence persists across the whole geometry grid we scanned rather than averaging out along it. Even the geometry-sensitivity hierarchy itself is ansatz-dependent once knob scales are accounted for [Fig. 4(4b)]. A geometry/trajectory optimization over the grid examined here therefore does not remove the coupling-induced ambiguity in the verdict. For this co-design decision the coupling must be constrained *before* the geometry search can be trusted. Orbital admixture, valley–orbit interaction, and  $g$ -factor anisotropy [35, 36, 39, 54] are the quantities that decide the outcome here. Existing literature addresses the relevant microscopic structure, and geometry remains an essential design variable. Our narrower, data-supported result is that the verdict is governed by the coupling model in a way the geometry grid does not resolve.

The value of this comparison is conditional on a specific fact: no generally validated effective local spin–valley kernel is currently available for the specific *shuttled-dot*/micromagnet regime considered here. Were such a model in hand one would simply adopt it and co-design directly. The family would collapse to the single true coupling, and the model-dependence we report would be moot. Design decisions today must instead be made under non-unique phenomenological proxies, so the disagreement between those proxies matters. A single proxy cannot expose it, however realistic-looking, because model dependence is visible only by *comparing* a family.

The atlas’s content is not merely that four deliberately different kernels produce four different numbers; that much is true by construction. The informative output is the *structure* of the disagreement. Magnitude orderings are largely conserved across the family while sign-level classifications are not, so the fragile quantity is the one a design decision consumes. The disagreement also concentrates on an identifiable physical axis, the gradient component modulated by localization, rather than being diffuse. The location and the asymmetry of this fragility do not follow from the kernels being different. The toy family is therefore the appropriate tool for a diagnostic of model sensitivity *under present ignorance*.

This model-risk result characterizes the current modeling problem for this reference geometry and diagnostic ansatz family: the thresholded classification, including its two-channel sign interpretation, is fragile across the family. Microscopically informed valley landscapes, such as those constrained by alloy-disorder calculations and conveyor-mode measurements [28, 34, 37], can be incorporated into the same atlas, and a microscopic spin-valley coupling can be added as a further member once one is validated for this setting. We expect them to narrow, but not by themselves collapse, the spread until one is actually validated for this setting.

*a. Limitations.* All conclusions are scoped to four toy-level effective coupling profiles, one fixed minimal spin-valley operator, one fixed charge-noise description, and the periodic micromagnet reference geometry examined here. The four ansätze differ only in their scalar spatial profiles, so operator-level alternatives are outside the present scope. The same  $1/f$  spectrum and a single representative, non-device-calibrated displacement-noise amplitude enter both responses, since  $\Delta P_v$  measures the effect of charge noise on valley leakage and  $\Delta\chi_\phi$  the effect of the valley sector on charge-noise dephasing. What is quantified here is therefore coupling-profile uncertainty conditional on that noise description; noise-spectrum and amplitude uncertainty are not scanned and form a separate axis.

The thresholded labels are operational classifications under the stated effect-size floors, not confidence-certified assignments. The threshold sweep and the targeted  $n_{\text{real}} = 30$  re-test address two distinct questions: sensitivity to the floor choice, and stability of the four screened candidates.

The robust/fragile statements above therefore apply to the four diagnostic ansätze and to the 432 model-conditions of the full validation grid (centred and edge pockets,  $v \in \{5, 10, 20\}$  m/s,  $n_{\text{real}} = 5$ ), executed as a per-case split and merged offline. Natural extensions are to measured or finite-device micromagnet fields, to families of noise descriptions, and to microscopically derived spin-valley couplings and operator structures; we leave these to future work because the purpose here is to isolate coupling-profile dependence against a fixed geometric backdrop.

*b. Implications for co-design.* Two practical implications follow. First, knob-importance claims depend on both the coupling ansatz and the normalization used for a geometry change, so no absolute statement survives scale accounting. Such claims should state which ansatz produced them and whether sensitivity is reported per absolute or per fractional variation. Second, because both the full thresholded classification and the two-channel sign interpretation are model-dependent and concentrated on the gradient component, any claim that a particular operating point is “robust” should be accompanied by a statement of which coupling model produced it. Proposed mitigations such as varying the shuttling velocity, dot shape, or path [29] should likewise be assessed across

coupling ansätze, or microscopic coupling models, rather than within one. The decisive next step is to calibrate the gradient structure of the synthetic coupling against a microscopic model or experiment, rather than to refine the geometry optimizer alone.

## VII. CONCLUSION

We have presented a sensitivity atlas for phenomenological spin-valley coupling ansätze in Si/SiGe shuttling. The atlas does not rank absolute shuttling performance; it ranks and classifies the baseline-relative response that the coupling induces, and asks how stable those rankings and classifications are across ansätze. The methodological contribution is twofold. First, a staged falsification chain plus a targeted higher-statistics re-test shows that a low-statistics robust classification can fail under increased Monte Carlo averaging. It also shows that whether a cross-ansatz robust classification survives that re-test is *convention-contingent*. The same procedure returns no surviving candidate in a stray-field Zeeman bookkeeping, and two grid-local robust survivors (edge pocket,  $v = 10$  m/s, reduced magnet-well separation, robust in all four ansätze at  $n_{\text{real}} = 30$ ) in the physically preferred total-field convention. This offers a template for guarding against premature “sweet-spot” claims in co-design studies more generally, while making explicit that the verdict itself carries modeling risk. Second, the atlas separates two notions of model dependence: the ordering of conditions and their physical classification. This localizes *where* modeling uncertainty matters most. Magnitude rankings are largely conserved across ansätze, while two-channel sign assignments are poorly conserved, particularly in the  $A-B_x$  and  $B_z-B_x$  comparisons. Co-design conclusions drawn from a single phenomenological  $H_{sv}$  ansatz and a single convention are therefore not trustworthy unless checked against alternative coupling descriptions and against absolute full-model error proxies.

The practical message for valley-aware co-design is therefore one of sequencing. In the model-uncertain setting examined here, geometry and trajectory optimization alone cannot settle the co-design verdict while the coupling form remains uncertain at the level we find here; effort is better spent first constraining the spin-valley coupling against microscopic theory or experiment. The atlas and its six metrics are designed to be reused for exactly this purpose. They accept additional coupling descriptions, including microscopically derived ones, as these become available, and report whether the spread we observe narrows.

*a. Future work.* The natural next step is to replace the phenomenological coupling ansatz with a microscopic effective  $H_{sv}$  derived from orbital admixture, valley-orbit interaction, and  $g$ -factor anisotropy [35, 36, 54]. Related gradient, strain, and spin-orbit design questions in Ge hole-spin platforms [40–42, 47, 48] can inform the same

program. Because our framework treats the coupling form as an interchangeable module, such a microscopic model can be added as an additional entry in the same atlas and compared pairwise against the toy families. We expect this to narrow the model dependence identified here. Residual uncertainty in microscopic input parameters will likely leave a finite degree of model dependence. Calibrating the gradient structure of the synthetic coupling against microscopic theory or experiment, rather than refining the optimizer, is the decisive next step toward quantitative valley-aware co-design.

## DATA AND CODE AVAILABILITY

The simulation code, the full-validate output tables, and the scripts that regenerate every figure and reported number are archived at Zenodo, DOI: [10.5281/zenodo.21588784](https://doi.org/10.5281/zenodo.21588784) (all versions: [10.5281/zenodo.21588003](https://doi.org/10.5281/zenodo.21588003)), and mirrored at [https://github.com/youinuk/spin-valley\\_model](https://github.com/youinuk/spin-valley_model). All quantitative results in this paper (rank correlations, classification agreements, and per-pair statistics) are reproducible from the released atlas outputs with a fixed random seed.

## USE OF AI TOOLS

Large-language-model assistants were used for editorial support, code review, and manuscript polishing. The

scientific claims, the simulation design and outputs, the figure data, and the final wording were checked and approved by the author, who is solely responsible for the content of this manuscript.

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## COMPETING INTERESTS

The author declares no competing interests.

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## ETHICS STATEMENT

This work is entirely theoretical and computational. It involved no human participants, animal subjects, or personal data, and required no ethical approval.

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